Muhammad Andrata Zharfan Mustika

AI Engineer | Data Scientist | Data Analyst
+62 822 3321 5051 | andrata2908@gmail.com | Bandung, West Java, Indonesia
linkedin.com/in/muhammad-andrata-zharfan-mustika/ | github.com/MAndrataZ

SUMMARY

Informatics Engineering graduate (S1) with a strong interest in AI and data, experienced in building LLM and RAG systems, machine learning for text and image data, and data analysis from preprocessing through evaluation. Comfortable leading teams, translating business requirements into technical solutions, and presenting results to technical and non-technical audiences. Seeking roles in AI engineering, data science, or data analysis.

EDUATION

Universitas Komputer Indonesia (UNIKOM), Bandung *2022 – 2026*
Bachelor's Degree (S1) in Informatics Engineering | GPA: 3,75 / 4,00

WORK & ORGANIZATIONAL EXPERIENCE

Fullstack Web Developer Intern | Rama Group (Rama FM & Paramuda Radio) *August 2025 - November 2025*

- Led development of the Rama FM and Paramuda Radio web platforms, covering system architecture, UI/UX improvements, and content management with Laravel, Inertia, and React.
- Served as the main technical liaison for stakeholders: coordinated the development team, managed schedules, and translated business requirements into technical solutions.
- Collaborated with the marketing division to position the website as a promotion and branding channel, and took part in client meetings.

General Secretary | Himpunan Mahasiswa Teknik Informatika (HMIF) UNIKOM, Informatics Engineering Student Association *2024 - 2025*

- Managed organizational administration: correspondence, meeting minutes, and activity documentation.
- Archived important documents (digital and physical) and supported the chair in coordinating work programs, internal policies, and meetings.

Badan Legislatif Mahasiswa UNIKOM (UNIKOM Student Legislative Body) *2023 - 2024*

- Channeled student aspirations for the Faculty of Engineering and Computer Science as a Member of Faction 1 FTIK in the Student Consultative Assembly (Majelis Permusyawaratan Mahasiswa, MPM).
- As a Member of Commission 5 (Creative Economy, Research, and IT) of the Student Representative Council (Dewan Perwakilan Mahasiswa, DPM), oversaw the performance of the ministry of creative economy, drafted policy, and managed DPM's social media and communications.

PROJECTS

Implementation of Retrieval-Augmented Generation (RAG) with a Qwen 2.5 7B Instruct Generator in a Stunting Question-Answering System (Undergraduate Thesis) *2025 - 2026*

Python · LLM · LangChain · RAG · RAGAS · FAISS · Gradio

- Built an Indonesian-language RAG pipeline that answers stunting questions from policy documents and scientific journals, using Qwen 2.5 7B Instruct, multilingual-e5-base embeddings, a FAISS vector index, and a Gradio demo application.
- Created a test dataset of 80 question-answer pairs across 6 question types, validated by two evaluators together with a nutrition expert using a human-in-the-loop approach.
- Evaluated with RAGAS (Context Precision 0.94; Answer Relevancy 0.90; Faithfulness 0.85) and retrieval MRR of 0.95; per-question analysis revealed a weakness on numerical questions (Faithfulness 0.69).

Early Detection of Heart Disease *2026*
Python · XGBoost · SHAP · MLJAR AutoML

- Built an early-detection classification model for heart disease on 908 patient records (11 features) with SHAP-based feature selection, then compared Decision Tree, Random Forest, Logistic Regression, and AutoML; Decision Tree (max_depth=3) performed best with 84.1% accuracy and 87.5% recall on the test set.

Child Stunting Status Classification from Basic Anthropometric Data *2026*
Python · Scikit-learn · Pandas · Seaborn

- Built a four-class classification of stunting status for children aged 0-24 months from four basic anthropometric variables on 92,692 records, comparing Logistic Regression (macro-F1 0.58) and Random Forest (macro-F1 of about 1.00), and critically assessing the near-perfect score because labels were derived from z-score rules on a synthetic dataset.

Sentiment Analysis of the Film Jumbo on Twitter (X) *2025*
Python · Scikit-learn · Pandas · Sastrawi · NLTK · imbalance-learn

- Built a three-class sentiment classification pipeline on 5,302 Indonesian-language tweets (CRISP-DM), covering preprocessing (slang normalization, Sastrawi stemming, negation handling), TF-IDF unigram-bigram features, and Logistic Regression.
- Addressed class imbalance (negative class only 5.4% of data) with class weights, SMOTE, RandomOverSampler, and 5-fold GridSearchCV: negative-class recall rose from 0.12 to 0.58, with 69% accuracy and macro-F1 of 0.63 on the test set.

Face Image Classification with CNN (ResNet50) *2025*
Python · PyTorch · Convolutional Neural Networks · OpenCV · Scikit-learn

- Built a three-class face image classification pipeline on 86,744 images, covering blur filtering with Laplacian variance (32,648 images remaining), face detection and cropping with Haar Cascade, data augmentation, and class imbalance handling with WeightedRandomSampler.
- Fine-tuned an ImageNet-pretrained ResNet50 in two stages, reaching 85.76% validation accuracy and macro-F1 of 0.85 (about 18 points higher than training the final layer only).

SKILLS

AI and Data: Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Prompt Engineering, Machine Learning, Deep Learning, Computer Vision, Sentiment Analysis, Model Evaluation (RAGAS), Data Preprocessing, Data Visualization.
Programming and Databases: Python, SQL, MySQL, JavaScript, PHP.
Framework and Tools: PyTorch, Scikit-learn, Hugging Face Transformers, LangChain, FAISS, Pandas, OpenCV, NLTK, Gradio, Laravel, React, Git, Jupyter Notebook.
Supporting: Team leadership and coordination, public speaking and effective communication, technical presentation, analytical and critical thinking, problem solving, objective evaluation.
Languages: Indonesian; English (TOEFL ITP: 550).

CERTIFICATIONS AND COURSE

Associate Data Scientist *2026*
Badan Nasional Sertifikasi Profesi (BNSP)
Gemini Certified University Student *2026*
Google
Artificial Intelligence & Automation Bootcamp *2026*
MySkill
AI for Work & Career Readiness With Google AI Products *2026*
Hacktiv8 Indonesia
Data Analyst Fundamental *2026*
MySkill
Google AI Essentials *2025*
Google
Software Engineer Fundamental Course *2024*
RevoU